Enhanced Adversarial Drift Detection (EADD) for MLOps Feature Monitoring

Abstract—**Machine learning models in production suffer from distribution drift, where evolving input features degrade model performance. Existing unsupervised detectors signal that a drift occurred, but not which features drifted or how severe the shift is. This paper proposes Enhanced Adversarial Drift Detection (EADD), extending the D3 discriminative detector with three contributions: (1) an Adaptive Sequential Permutation Test (ASPT) for statistically rigorous drift confirmation, reducing the permutation budget by 90–95%, with prediction entropy emitted as a continuous diagnostic output (not evaluated standalone); (2) SHAP-based drift localization and a Drift Severity Index (DSI) for feature-level diagnosis and severity quantification; and (3) experiments benchmarking against seven state-of-the-art unsupervised detectors on synthetic and real-world streams. On synthetic data, EADD is the only detector achieving full drift-type coverage (4/4) with zero false alarms. On real-world benchmarks, EADD detects drift on three of four annotated datasets, outperforming D3 on detection delay in all three detected cases, up to 8× faster on incremental drift types (InsectsIncrAbrupt MTD 33 vs. D3 272.5). SHAP attribution achieves perfect accuracy (P@k = R@k = 1.0) on controlled synthetic scenarios with independent features. On stable streams, EADD produces zero false alarms versus D3's 54.6 on autocorrelated streams at the matched threshold of 0.7; the reduction versus D3 at threshold 0.6 is statistically significant ($p = 0.0101$, Mann–Whitney U).**

Index Terms—**concept drift, discriminative classification, feature attribution, distribution shift, MLOps, drift localisation**

I. Introduction

Machine learning models deployed in production environments encounter evolving data distributions, collectively termed distribution drift, that can degrade model performance if left undetected [1], [2]. Drift may manifest as changes in input features (covariate drift) or in the input–output relationship (concept drift); detecting such changes early is essential for maintaining reliable inference [3], [4].

The drift detection landscape spans three paradigms. *Supervised methods* such as DDM [8] monitor error rates but require timely ground-truth labels. *Statistical methods* employ hypothesis tests such as the Kolmogorov–Smirnov test [6], [7] but operate per feature, missing multivariate shifts. *Discriminative methods* such as D3 [9] train classifiers to distinguish reference from current data, achieving strong robustness in the benchmark by Lukats et al. [10]. However, all existing methods share a common limitation: they report only *whether* drift occurred, not *which* features drifted or *how severe* the shift is.

D3 [9], among the most robust unsupervised detectors in the Lukats et al. benchmark [10], exemplifies this limitation with two gaps. First, D3 applies no formal significance test, relying instead on a fixed AUC threshold without statistical justification, which makes it vulnerable to false alarms on autocorrelated data. Second, D3 follows a "detect-and-discard" pattern: the trained classifier is discarded after computing AUC, providing no information about which features drove the shift. Recent work has partially addressed related gaps: TRIPODD [27] enables feature-interaction hypothesis testing but requires pre-specifying interactions and has quadratic complexity; HDPD [28] provides per-feature performance attributions but requires labels and operates post-hoc on collected batches.

In the *academic research* literature on unsupervised streaming detectors, no single framework integrates (1) statistically rigorous multivariate detection, (2) identification of *which* features caused the shift, and (3) quantification of *how severe* the drift is. Commercial monitoring platforms (e.g., Vertex AI [20], SageMaker Clarify [21]) address aspects of this in production but operate in supervised or batch settings and are not designed for real-time streaming. Nor has any prior work systematically evaluated the accuracy of SHAP-based feature attribution within discriminative streaming drift detection.

EADD is therefore an *integration contribution*: each component draws on established methods (Gandy [29] for sequential testing, TreeSHAP [24], [30] for attribution), but their combination within a single unsupervised streaming pipeline is non-trivial. The AUC gate must precede ASPT to bound overall Type I error conservatively; TreeSHAP must be applied to a *discriminative* rather than predictive classifier; and the severity metric must be defined dimensionality-invariant to handle varying feature counts across datasets.

This paper proposes **Enhanced Adversarial Drift Detection (EADD)**, addressing this gap with three contributions:

1) Inspired by the statistical analysis of the classifier two-sample test [19], [26], we introduce an **Adaptive Sequential Permutation Test (ASPT)** [29] that gates drift confirmation via a formal permutation test, achieving 90–95% permutation savings while preserving Type I error control; H_{pred} is emitted as a continuous diagnostic output at every monitoring step (Section III-B,C).
2) Building on SHAP feature distribution analysis [24], we propose **SHAP-based drift localisation** and a **Drift Severity Index (DSI)** that combines detection confidence with attribution concentration into a continuous severity score, identifying which features caused drift and quantifying how severe the shift is (Section III-D).
3) We benchmark EADD against seven unsupervised detectors on synthetic and real-world streams, evaluating detection coverage, false alarm control, and SHAP attribution

accuracy (Section IV).

II. Related Work

A. Drift Detection Methods

Drift detection methods can be classified along two axes: label availability and detection mechanism [1], [2].

Supervised methods monitor model error rates and require timely ground-truth labels. DDM [8] tracks the error rate and its standard deviation; EDDM [11] monitors inter-error distances, improving sensitivity to gradual drift. Both react only *after* performance has degraded, limiting utility when labels are delayed [1], [5].

Unsupervised methods operate without labels by directly monitoring data distributions. Statistical approaches include ADWIN [12], which dynamically adjusts sliding window sizes and can detect gradual drift, and KSWIN [7], which applies the Kolmogorov–Smirnov two-sample test over sliding windows for abrupt or sudden shifts. While statistically principled, these methods test each feature *independently* (univariate), making them blind to multivariate distributional shifts and causing multiple testing problems in high-dimensional settings [6].

Discriminative approaches frame drift detection as binary classification: an auxiliary classifier is trained to distinguish reference from current distributions [9], [13], [14], [19]. This naturally captures multivariate drift in a single test. The most prominent discriminative detector, **D3** [9], trains a logistic regression classifier on reference vs. current windows and triggers drift when AUC exceeds a fixed threshold. Lukats et al. [10] identified D3 as among the most robust unsupervised detectors across 29 real-world data streams. However, D3 has three weaknesses: (1) its linear discriminator misses non-linear feature interactions, causing failure on gradual and incremental drift; (2) it applies no formal significance test, relying on a fixed AUC threshold vulnerable to false alarms on autocorrelated data; and (3) it discards the trained classifier after computing AUC, providing no feature-level attribution. Other benchmarked detectors exhibit characteristic limitations: CSDDM [15] and IBDD [16] achieve broad coverage but with elevated false alarm rates; OCDD [17] and SPLL [18] are effective for specific drift types but miss others; BNDM struggles with high-dimensional data; and UDetect failed on 45% of scenarios [10].

B. Feature Attribution in Discriminative Detection

The *classifier two-sample test* (C2ST) [19] provides principled statistical foundations for discriminative detection: if a classifier reliably separates reference from current data (AUC $\gg$ 0.5), the distributions differ significantly. E-C2ST [26] strengthened statistical rigour via anytime-valid e-values. However, neither C2ST nor E-C2ST provides feature-level attribution or operates in a streaming setting. The practical need for such attribution was demonstrated at industrial scale by Pan et al. [13] at Uber, Qian et al. [14] for credit scoring, and by Google's SHAP-based feature monitoring in Vertex AI [20].

Recent academic work has advanced individual aspects: TRIPODD [27] enables feature-interaction hypothesis testing, HDPD [28] provides per-feature performance decomposition, and Hinder et al. [31] formalised feature-wise drift definitions. However, no existing work systematically evaluates SHAP-based feature attribution accuracy for identifying drift-causing features within discriminative detection, nor integrates attribution with severity quantification in an unsupervised streaming pipeline.

III. Proposed Methodology

As identified in Section II, existing C2ST-based detectors lack continuous uncertainty monitoring, statistical significance testing, feature-level attribution, and severity quantification. EADD is a streaming pipeline for feature drift ($P(X)$ shift) detection that extends the C2ST framework [19] with three novel components to address these gaps (Fig. 1). The complete detection cycle is specified in Algorithm 1, which is referenced throughout this section; the four subsections that follow expand each step in turn.

Algorithm 1 EADD Detection Cycle

Require: Stream $\{x_t\}$, W_{ref}, W_{cur}, threshold τ, significance α

Ensure: Drift decision, H_{pred}, DSI, feature attribution, prescription

```
Update W_cur; update W_ref via reservoir sampling [22]
Train LightGBM on W_ref ∪ W_cur with labels {0, 1}
Compute AUC via stratified 5-fold CV
Compute H_pred via Eq. (1)
if AUC > τ then
    p ← ASPT(W_ref, W_cur, AUC) {Adaptive early stopping}
    if p < α then
        Compute SHAP values via TreeSHAP
        Compute DSI via Eq. (2)
        Generate DSI-based prescription
        return DRIFT, H_pred, DSI, SHAP ranking, prescription
    end if
end if
return NO-DRIFT, H_pred
```

A. Step 1: Adaptive Reference Windowing

EADD maintains two windows: a *reference window* W_{ref} (N_{ref} samples) and a *current window* W_{cur} (N_{cur} samples). The reference window is updated by *reservoir sampling* [22]: each new sample x_t replaces a random element in W_{ref} with probability N_{ref}/t, which keeps an unbiased sample of historical data over long streams. The current window is a standard sliding buffer of recent observations. Monitoring is triggered every M samples to balance latency and cost; after confirmed drift, W_{ref} is reset to the new regime.

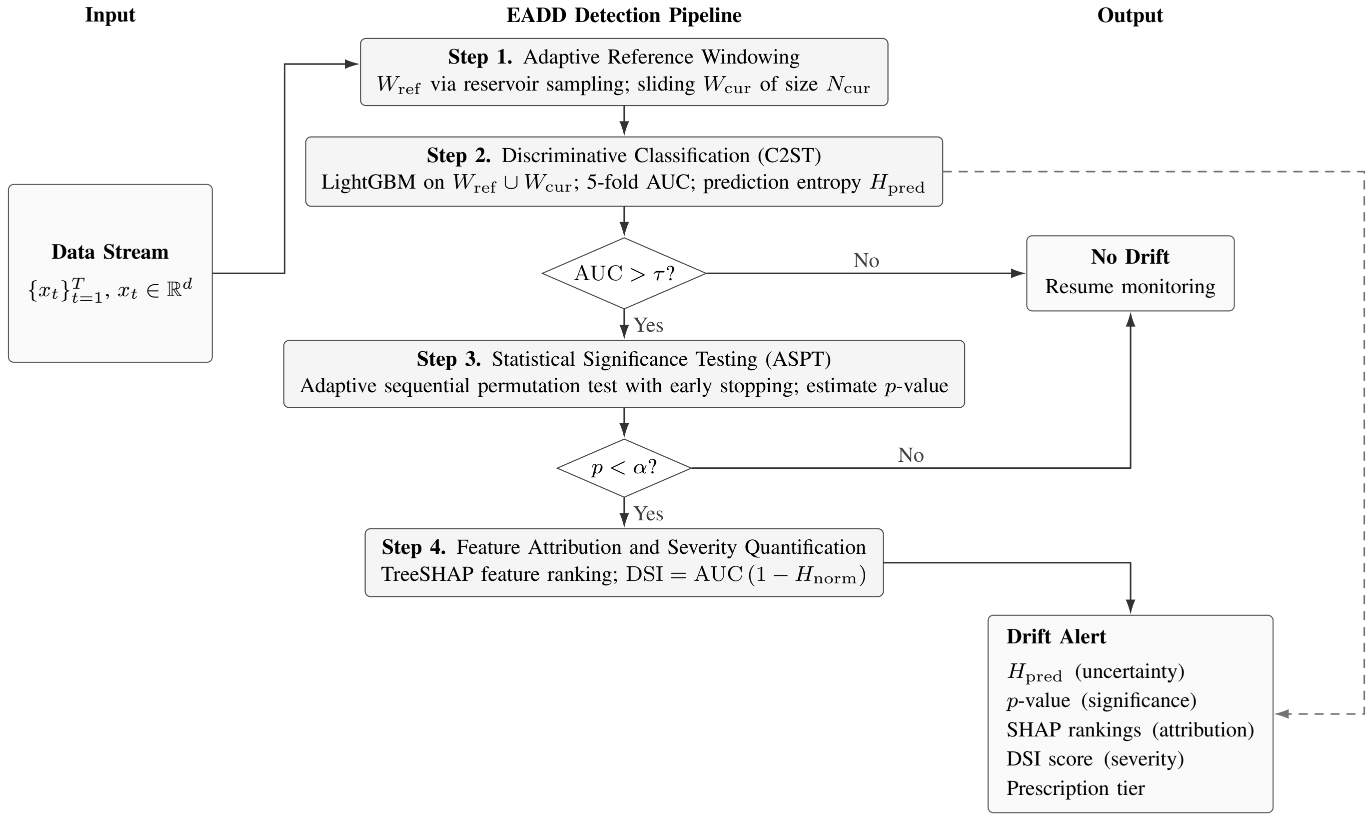

Fig. 1. EADD detection pipeline. W_{cur} is compared against W_{ref} via a discriminative classifier; H_{pred} provides a continuous uncertainty monitoring signal (dashed). AUC $> \tau$ triggers ASPT for statistical confirmation (Step 3); upon significance, TreeSHAP and DSI localise and quantify the shift (Step 4).

B. Step 2: Discriminative Classification and Prediction Entropy

Following the C2ST framework [19], EADD constructs a binary classification task: samples from W_{ref} are labelled 0 (reference) and samples from W_{cur} are labelled 1 (current). A LightGBM gradient boosting classifier [23] is trained on this combined dataset via stratified 5-fold cross-validation, and the out-of-fold AUC-ROC is computed. AUC $\approx$ 0.5 indicates the classifier cannot distinguish the two windows (no drift); AUC $\gg$ 0.5 indicates the distributions are separable (drift suspected). We use LightGBM rather than D3's logistic regression [9] because gradient boosting captures non-linear feature interactions, providing a more powerful discriminator for the non-linear and incremental drift types that defeat D3's linear classifier.

Prediction Entropy. Beyond the aggregate AUC score, EADD extracts additional diagnostic information from the classifier's per-sample predictions. We compute the mean binary Shannon entropy of the discriminative classifier's predicted probabilities:

$$H_{\text{pred}} = \frac{-1}{n} \sum_{i=1}^{n} \left[\hat{p}_i \ln \hat{p}_i + (1-\hat{p}_i) \ln(1-\hat{p}_i)\right] \qquad (1)$$

where $\hat{p}_i = f_\theta(x_i)$ is the probability that sample i belongs to W_{cur}: $H_{\text{pred}} \approx \ln 2 \approx 0.693$ indicates maximum uncertainty (no drift); $H_{\text{pred}} \to 0$ indicates high classifier confidence (drift likely). Note: H_{pred} uses natural log; its maximum is $\ln 2$, not 1. H_{pred} remains informative at every monitoring step, including steps where AUC $< \tau$ and no detection gate is triggered, providing a continuous sub-threshold trend signal; its standalone predictive value is qualitative in the present work.

C. Step 3: Adaptive Sequential Permutation Test (ASPT)

If AUC exceeds a threshold τ, EADD applies an **adaptive sequential permutation test** inspired by Gandy [29]. Instead of a fixed budget of B permutations, ASPT monitors evidence accumulation and terminates early:

1) For each permutation $i = 1, \ldots, B_{\max}$: shuffle labels, retrain, compute $\text{AUC}^{(i)}_{\text{perm}}$. Update count $c_i = \#\{j \le i : \text{AUC}^{(j)}_{\text{perm}} \ge \text{AUC}_{\text{actual}}\}$.
2) **Early reject H_0:** If $c_i = 0$, $i \ge B_{\min}$, and $1/(i+1) < \alpha$, terminate and confirm drift with $p = 1/(i+1)$.
3) **Early accept H_0:** If $c_i/i > 2\alpha$ and $i \ge B_{\min}$, terminate and declare no drift with $p = (c_i + 1)/(i + 1)$.
4) **Terminal:** If $B_{\max}$ is reached, $p = (c_{B_{\max}}+1)/(B_{\max}+1)$.

Drift is confirmed if $p < \alpha$. The two-stage gate is conservative: overall Type I error $\le \Pr(\text{AUC}{>}\tau \mid H_0) \times \alpha \le \alpha$. Under $\alpha = 0.05$, early rejection requires $i \ge 20$ (since $1/21 \approx$

$0.048 < 0.05$); all confirmed detections in Experiments 1 and 3 terminated at exactly $i = 20$.

D. Step 4: Feature Attribution and Drift Severity Index

Upon drift confirmation, EADD exploits the trained discriminative classifier to diagnose *which* features drive the detected drift using SHAP (SHapley Additive exPlanations) [24].

Feature Attribution. EADD applies TreeSHAP [30] to the trained LightGBM discriminator. Per-feature importances $\bar{\phi}_j = \frac{1}{n}\sum_{i=1}^{n}|\phi_j^{(i)}|$ are normalised into a contribution vector $\mathbf{s} = (s_1, \ldots, s_d)$, $s_j = \bar{\phi}_j / \sum_k \bar{\phi}_k$, identifying which features drive the shift and feeding the DSI below.

Drift Severity Index (DSI). The normalised contribution vector s computed from TreeSHAP directly feeds the DSI: a composite severity metric combining detection confidence with attribution concentration:

$$\text{DSI} = \text{AUC} \times (1 - H_{\text{norm}}(\mathbf{s})) \tag{2}$$

where $H_{\text{norm}}(\mathbf{s}) = -\sum_{j=1}^{d} s_j \log_2 s_j \,/\, \log_2 d$ is the normalised Shannon entropy [33]. The $\log_2 d$ denominator ensures $H_{\text{norm}} \in [0, 1]$ regardless of dimensionality. DSI-based prescriptions map severity to recommended action. The tier boundaries below are proposed heuristics requiring domain-specific calibration; they are not empirically validated in the present work (all experiment DSI values fall in the Low tier):

- **Critical (DSI > 0.6):** Concentrated drift, audit specific data pipeline.
- **Moderate (0.3–0.6):** Subset drift, investigate shared data source.
- **Low (< 0.3):** Diffuse shift, schedule full model retraining.

IV. Experimental Evaluation

A. Experimental Design

Three experiments were designed to validate EADD's detection accuracy, feature attribution, and computational efficiency. Following the evaluation protocol of Lukats et al. [10], EADD was benchmarked against seven state-of-the-art unsupervised detectors: D3 [9], BNDM, CSDDM [15], IBDD [16], OCDD [17], SPLL [18], and UDetect.

Datasets. Table I summarises the datasets used across experiments.

Experiment 1 (Synthetic Data): Synthetic streams (n=5,000, d=5) with controlled abrupt, gradual, incremental, and recurring drift. All eight detectors were compared on each drift type. Additionally, false alarm robustness was evaluated on four stable synthetic streams (Gaussian i.i.d., autocorrelated AR(1) with ϕ=0.8, heteroscedastic, correlated with ρ=0.7) containing zero drift, with five independent runs per stream type.

Experiment 2 (Real-World Benchmark): Thirteen datasets (Table I): four INSECTS variants with annotated drift points [10] are evaluated by Recall and Mean Time to Detection (MTD); nine unannotated streams [25] are evaluated by total detection count (selectivity). Only EADD and D3 are compared in this experiment.

TABLE I
Datasets used in experiments (†=annotated; drift count in parentheses; Unannotated = no ground-truth drift-point labels).

Dataset	n	d	**Drift?**
Synthetic (Experiments 1, 3)			
Temporal Patterns	5K	5	Yes (4 types)
Feature Attribution	10K	10	Yes (3 scenarios)
Experiment 2: Real-World [10], [25]			
INSECTS Abrupt Bal.†	52.8K	33	Yes (5)
INSECTS Gradual Bal.†	24.2K	33	Yes (1)
INSECTS IncrAbrupt Bal.†	80.0K	33	Yes (2)
INSECTS IncrRecur Bal.†	80.0K	33	Yes (2)
Electricity	45.3K	8	Unannotated
INSECTS Incremental	57.0K	33	Unannotated
Luxembourg	1.9K	31	Unannotated
NOAAWeather	18.2K	8	Unannotated
OutdoorObjects	4.0K	21	Unannotated
Ozone	2.5K	72	Unannotated
PokerHand	100K	10	Unannotated
Powersupply	29.9K	2	Unannotated
RialtoBridge	82.3K	27	Unannotated

Experiment 3 (SHAP Attribution Accuracy): Three controlled synthetic scenarios (n=10,000, d=10 features, drift injected at t=5,000): (a) univariate drift in F3 only (mean shift $+2\sigma$), (b) subset drift in F2/F5/F7 simultaneously, (c) multivariate drift in all features. Precision@k (P@k) and Recall@k (R@k) measure whether the top-k SHAP-ranked features exactly match the k truly drifted features.

Configuration. EADD uses a LightGBM [23] discriminator with 50 estimators, a learning rate of 0.1, and 31 leaves, chosen to balance classifier capacity against the small per-window training budget. The reference window holds N_{ref}=500 samples and the sliding current window holds N_{cur}=200 samples, checked every M=50 stream samples to bound monitoring latency. The ASPT stage is configured with a maximum of $B_{\max}$=199 permutations and a minimum of $B_{\min}$=10 to enable early stopping, while the significance level α=0.05 and AUC gate τ=0.7 match D3's conservative operating point for a fair comparison. For the baseline, D3 [9] retains its original logistic regression classifier (liblinear solver) and uses the same window sizes and fixed AUC threshold of 0.7. All experiments run under Python 3.10 with LightGBM 4.1.0, SHAP [24] 0.42.1, and scikit-learn 1.3.2; random seeds are fixed across detectors to ensure reproducibility.

B. Experiment 1: Synthetic Data

Table II presents drift type coverage and false alarm rates for all eight detectors.

EADD is the only detector that achieved full drift-type coverage (4/4) with zero false alarms. We attribute the coverage advantage to LightGBM's non-linear discrimination, consistent with D3's documented failure on gradual and incremental drift [9], [10], though no ablation isolates this from other design choices (see Section VI). The ASPT then eliminates spurious detections on stable streams, bringing false alarms to zero. BNDM, CSDDM, IBDD, and OCDD also

TABLE II
EXPERIMENT 1: SYNTHETIC DRIFT DETECTION RESULTS. TYPES = NUMBER OF DRIFT PATTERNS DETECTED OUT OF 4. MEAN FA = AVERAGE FALSE ALARMS PER TYPE. LIFT = DETECTIONS PER TRUE DRIFT POINT, FOLLOWING LUKATS ET AL. [10] REPORTING.

Detector	Types	Coverage	Mean FA	Lift
EADD	**4/4**	**100%**	**0.00**	**1.00**
BNDM	4/4	100%	4.75	5.75
CSDDM	4/4	100%	5.50	6.50
IBDD	4/4	100%	46.25	47.25
OCDD	4/4	100%	17.50	18.50
D3	2/4	50%	0.00	1.00
SPLL	2/4	50%	0.00	1.00
UDetect	0/4	0%	n/a	n/a

achieved 4/4 coverage (through different discriminators) but with substantially higher false alarm costs (4.75, 5.50, 46.25, and 17.50 mean FA, respectively), demonstrating that coverage and false alarm control are separable objectives. D3 and SPLL detected only 2/4 types. UDetect failed on all four types.

False Alarm Robustness. On four stable streams, EADD produced zero false alarms. D3 was vulnerable to autocorrelated data (87.4 false alarms at τ=0.6; 54.6 at τ=0.7); ASPT correctly identifies autocorrelation as non-significant ($p > 0.05$). A Mann–Whitney U test confirmed significantly fewer false alarms for EADD vs. D3 at τ=0.6 ($p = 0.0101$, D3's benchmark default [10]); at the matched τ=0.7 operating point, the difference is not statistically significant ($p = 0.2266$), though EADD still produces zero false alarms vs. D3's 54.6 on autocorrelated streams.

C. Experiment 2: Real-World Benchmark

Table III presents results on the four annotated INSECTS datasets. Both detectors missed InsectsGradual, a slow transition detectable only at higher false alarm cost [10]. On all three jointly detected datasets, EADD outperforms D3 on MTD, most markedly on incremental drift types: InsectsIncrAbrupt MTD 33 vs. 272.5 (8× faster) and InsectsIncrRecur MTD 133 vs. 323.5 (2.4× faster). On InsectsAbrupt, EADD achieves MTD 135 vs. 155.5 (13% faster) with lower recall (0.60 vs. 0.80), reflecting its precision-first operating point. All MTD values are single-run point estimates. On nine unannotated datasets, EADD and D3 produced identical total detection counts (467 each).

D. Experiment 3: SHAP Attribution Accuracy

Table IV presents the SHAP attribution accuracy using precision@k and recall@k metrics.

EADD achieved perfect attribution accuracy across all three scenarios; the DSI-based prescription was correct in all cases. The multivariate scenario (k=d=10) trivially satisfies P@k = R@k = 1.0 for any ranking, so the meaningful result is the univariate case (1/10 features drifted, correct feature F3 ranked first at 52.3%). DSI correctly orders scenarios by concentration (univariate > subset > multivariate), as algebraically expected from Eq. (2). All DSI values fall in the "Low" tier (< 0.3) under moderate $+2\sigma$ shifts; Critical and Moderate tiers require more severe drift and are not empirically reached here.

TABLE III
EXPERIMENT 2: ANNOTATED REAL-WORLD DATASETS. PREC. = PRECISION; REC. = RECALL ON ANNOTATED DRIFT POINTS. MTD = MEAN TIME TO DETECTION (SAMPLES; LOWER IS BETTER). "N/A" = MISSED. LOW PRECISION ON INCREMENTAL STREAMS REFLECTS SPARSE ANNOTATION OF GRADUAL TRANSITIONS [10].

Dataset	Metric	EADD	D3
InsectsAbrupt	Precision	0.33	0.44
InsectsAbrupt	Recall	0.60	0.80
InsectsAbrupt	MTD	**135**	155.5
InsectsGradual	Precision	0.00	0.00
InsectsGradual	Recall	0.00	0.00
InsectsGradual	MTD	n/a	n/a
InsectsIncrAbrupt	Precision	0.14	0.22
InsectsIncrAbrupt	Recall	**1.00**	1.00
InsectsIncrAbrupt	MTD	**33**	272.5
InsectsIncrRecur	Precision	0.17	0.18
InsectsIncrRecur	Recall	**1.00**	1.00
InsectsIncrRecur	MTD	**133**	323.5

TABLE IV
EXPERIMENT 3: SHAP ATTRIBUTION ACCURACY ON CONTROLLED SYNTHETIC STREAMS (n=10,000, d=10 FEATURES; DRIFT INJECTED AT t=5,000 IN F3 ONLY / F2+F5+F7 / ALL FEATURES). ALL SCENARIOS DETECTED AT STEP 5,149. DSI = AUC × $(1 - H_{\text{NORM}}(\mathbf{s}))$; HIGHER DSI INDICATES MORE CONCENTRATED, SEVERE DRIFT. RX (✓) = CORRECT DSI-BASED DRIFT-TYPE CLASSIFICATION.

Scenario	AUC	P@k	R@k	Top (%)	DSI	Rx
Univariate (1/10)	0.784	1.0	1.0	F3 (52.3)	0.193	✓
Subset (3/10)	0.809	1.0	1.0	F5 (24.5)	0.096	✓
Multivariate (10/10)	0.806	1.0	1.0	F2 (14.6)	0.016	✓

V. CONCLUSION

This paper presented EADD, extending discriminative drift detection with (1) ASPT + H_{pred} for statistically rigorous drift confirmation, (2) SHAP attribution and DSI for feature-level diagnosis and severity quantification, and (3) evaluation against seven state-of-the-art detectors on synthetic and real-world streams.

Evaluation demonstrated that EADD achieves full drift-type coverage (4/4) via LightGBM's non-linear discrimination, with zero false alarms via ASPT's significance filter, the only detector to achieve both simultaneously on synthetic benchmarks. EADD detects drift on three of four annotated real-world datasets, outperforming D3 on MTD in all three detected cases (InsectsAbrupt: 135 vs. 155.5; InsectsIncrAbrupt: 33 vs. 272.5, 8× faster; InsectsIncrRecur: 133 vs. 323.5; single-run estimates without variance). SHAP attribution achieves perfect accuracy (P@k = R@k = 1.0) on controlled synthetic scenarios with independent features; the non-trivial result is univariate attribution (1/10 drifted features correctly identified). On stable streams, EADD produces zero false alarms vs. D3's 54.6 on autocorrelated streams at the matched τ=0.7 operating point; the reduction vs. D3 at τ=0.6 is statistically significant ($p = 0.0101$). By answering "Is drift emerging?", "Is it significant?", "Which features caused it?", and "How severe?", EADD extends binary drift alarms with feature-level attribution and severity scoring.

VI. Limitations and Future Work

ASPT reduces the permutation budget by 90–95% relative to fixed-budget ($B_{\max}$=199) permutation testing, a saving within EADD's own statistical stage. Separately, the discriminative paradigm overall remains $\sim$50$\times$ costlier than D3; this overhead arises from LightGBM training, not the permutation test, and is independent of the ASPT saving. No runtime table is reported. DSI ordering across scenarios is algebraically expected from its formula and does not constitute independent empirical validation; external ground-truth validity is not established, as we do not show that higher DSI correlates with downstream model performance degradation. SHAP attributions are correlational, not causal [32]. Attribution accuracy was tested on 3 controlled synthetic scenarios with independent features; performance under correlated or redundant features requires separate validation. No ablation isolates component contributions: replacing D3's logistic regression with LightGBM alone (without ASPT) may already achieve 4/4 coverage, so the benefit attributable to ASPT specifically is unknown. DSI tiers require domain-specific calibration. Prescriptions are heuristic and not evaluated by downstream outcomes (e.g., retraining quality). **Multiple testing over time**: the pipeline runs a hypothesis test every M=50 samples with $\sim$75% window overlap, each at α=0.05 and no correction applied across time steps. The family-wise error rate therefore inflates with stream length; the zero false alarm result on synthetic streams is empirical rather than theoretically guaranteed. Sequential testing with anytime-valid e-values [26] is a natural candidate to address this gap. Future work: e-value testing [26], causal attribution [32], adversarial robustness.

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